Vel Tech Group of Institutions

Name: GOPICHAND KAKIVAI
Email: vtu29748@veltech.edu.in
Roll no: vtu29748
Phone: null
Branch: VTU
Department: CSE
Batch: 2028
Degree: B.Tech - Computer Science and Engineering

Scan to verify results

2028_NeoColab_Design and Analysis of AlgorithmsS1L3

VTU_DAA_Week 2_COD

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Given an integer n. The task is to find the first n Fibonacci Numbers.

Input Format

The input consists of a single integer n, representing how many Fibonacci terms to print.

Output Format

The output prints n space-separated Fibonacci numbers starting from 0.

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: 3
Output: 0 1 1

Answer

```
#include<stdio.h>
int main(){
   int n;
   scanf("%d",&n);
   int a=0,b=1, next;
   for(int i=0;i<n;i++){
      printf("%d",a);
      next=a+b;
      a=b;
      b=next;
   }
   return 0;
}
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

Given a string s, check if it is a palindrome using recursion. A palindrome is a word, phrase, or sequence that reads the same backward as forward.

Input Format

The input consists of a single string s containing no spaces.

Output Format

The output prints "The string is a palindrome" if the string is symmetric.

Otherwise, prints "The string is not a palindrome".

Refer to the sample output for the formatting specifications.

Sample Test Case

Input: abba
Output: The string is a palindrome

Answer

```
#include<stdio.h>
#include<string.h>
int ispalindrome(char s[],int start,int end){
   if(start>=end){
      return 1;
   }
   if(s[start]!=s[end]){
      return 0;
   }
   return ispalindrome(s, start+1,end-1);
}
int main(){
   char s[51];
   scanf("%s",s);
   int len=strlen(s);
   if(ispalindrome(s,0,len-1)){
      printf("The string is a palindrome");
   }else{
      printf("The string is not a palindrome");
   }
   return 0;
}
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Shilpa is learning about the Fibonacci sequence for her computer science class. She wants to generate the first N terms of the Fibonacci series.

Can you help Shilpa by writing a program that takes a number N and prints the first N terms of the Fibonacci series using recursion?

Input Format

The input consists of an integer N, representing the number of terms in the Fibonacci sequence that need to be printed.

Output Format

The output prints the first N Fibonacci numbers, separated by spaces.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6
Output: 0 1 1 2 3 5

Answer

```
import java.util.Scanner;

#include<stdio.h>
int fib(int n){
   if(n==0)
   return 0;
   else if(n==1)
   return 1;
   return fib(n-1)+fib(n-2);

}

public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int N = sc.nextInt();
      for(int i = 0; i < N; i++) {
         System.out.print(fib(i));
         if(i != N - 1) System.out.print(" ");
      }
      System.out.println();
   }
}
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

Ashish is preparing a mathematics assignment where he needs to calculate the factorial of a number. Since his teacher asked him to use recursion for this task, he plans to complete the method int factorial to solve it.

Help him to implement the task.

Input Format

The input consists of an integer num representing the number for which the factorial is to be calculated.

Output Format

The output prints the factorial value of num.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 3

Output: 6

Answer

```
#include <stdio.h>

#include<stdio.h>
int factorial(int num){
   if(num==0||num==1){
      return 1;
   }else{
      return num*factorial(num-1);
   }
}


int main() {
   int num;
   scanf("%d", &num);
   printf("%d\n", factorial(num));
   return 0;
}
```

Status : Correct *Marks : 10/10*